

and many more possibilities. Using power-line communication (PLC) or other wired and wireless technologies like low-power radio frequency to connect the meter to the service provider enables all of the above features to be feasible and compatible with future smart-grid protocols," explains Mongia.

"Smart metering also provides a way to offer increasingly personalised services and reduce pollution and carbon emissions," adds Naveen Chopra, director, Vodafone Business Services, a leading provider of communication services for smart grids.

STMicroelectronics offers a complete set of semiconductor products to implement energy, water, heat and gas meters. These products include concentrators, multi-utility controllers, microcontroller units for wireless sensor networks, low-power wireless transceivers and so on. Their STaGRID platform also helps in energy and grid control applications such as smart grids, smart metering, smart building/home and e-mobility applications. The platform includes various SoC implementations (ST7570, ST7580 and ST7590) featuring high modularity and flexibility, multiple modulations and protocols, and high integration (the DSP PHY processor, protocol stack engine, analogue front end, power-line driver, encryption and auxiliary functions are packaged in a single chip). It complies with major protocol specifications such as IEC 61334-5-1, PRIME, Meters and More, and others.

Communications backbone for smart grids

"Smart grid architecture requires an intense communication network to transmit the data and enable utilities to take up automated actions based on the information generated from the real-time data. Basically, in the smart grid you are monitoring, controlling and managing the demand and supply of the energy. In order to do this, you need to know on a real-time basis the level of energy consumption by users. At the same time, you need to know supply of energy. All of this requires

a robust communication network as the backbone for exchange of information between various systems," says Chopra of Vodafone.

To explain this better, Chopra cites the example of a manufacturing unit connected to the smart grid. Suppose at some point of time, the utility provider faces a shortage of electricity supply in one of the regions. With smart grid technology, the utility has complete information about the energy consumption pattern of this manufacturing unit, so it can offer a discount to the manufacturing unit for reducing energy consumption during that time period, and remotely control the supply to this manufacturing unit. Now this reduction in the consumption of energy will enable the utility to divert the extra available electricity to the area that requires it the most, hence controlling the demand and supply on a real-time basis.

Demand and supply management works only if you have complete online information about each and every distribution point, including details about the level of consumption at every point. This is possible only if all the consumption points are smart (let's say, with smart meters) and able to communicate. This makes communication a key infrastructure for implementing smart grids.

A smart grid requires high-availability, secure, two-way external communication that is resilient and resistant to external interference. It should not be affected by interference from other devices in the home or exceed electro-sensitivity standards. It should have the capacity to handle the anticipated volume and resource demands of smart metering.

The network infrastructure would typically be hybrid, with both wireless and fixed-line communications. Consider the example of a Vodafone implementation. Each remote meter is made smarter by enabling communication using the Vodafone GSM network. These meters can communicate data to regional centres and applications, which are connected using a Vodafone fixed-line data network. Further, regional

servers and applications are connected to the central data centre over a Vodafone MPLS network. In addition, the utility provider's control centres can use Vodafone Internet leased line and fixed-line voice to connect to the public infrastructure at large.

M2M communication in energy management

Machine-to-machine (M2M) communication plays a key role in energy management across segments and applications.

"For example, a smart grid can use innovative M2M communication to create a single integrated service that monitors and controls flows, ensures the safety of the smart grid network and delivers the peak demand when necessary. M2M can make homes, buildings and industrial units smarter by allowing the consumption of services such as energy to be closely monitored and managed more effectively over their lifecycle. It also enables energy and utility providers to efficiently collect meter data and provide two-way communication and tailored offers for their residential, commercial and industrial customers," says Chopra.

Vodafone provides a range of M2M solutions for energy management, including managed connectivity, powered by the M2M service platform, private APN services for secured connectivity, private static IP solution and special industry-grade SIM cards. These solutions are already used by utilities for smart metering as part of their automation programmes and as part of the central government driven R-APDRP programme.

"M2M solutions are helping utilities to provide a secure and sustainable future for all," adds Chopra.

Solar to the rescue

Traditional sources of energy are fast depleting, and solar is rising to the occasion as a brilliant alternative. There are industrial as well as consumer-level products and solutions in this space, and their efficiency and costs are driven largely by developments in the